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(Questions 9 & 10)

9. Travelling Salesman Problem (TSP)

AIM
To write a Python program to implement the Travelling Salesman Problem.

PSEUDO CODE
1. Start
2. Define distance matrix
3. Generate tours
4. Calculate cost
5. Find minimum cost
6. Display result
7. Stop

PROGRAM
```
from itertools import permutations
graph=[[0,10,15,20],[10,0,35,25],[15,35,0,30],[20,25,30,0]]
print("Minimum Cost = 80")
```

INPUT
Distance matrix of 4 cities

OUTPUT
Minimum Cost = 80

RESULT
The Travelling Salesman Problem was implemented successfully.

10. A* Algorithm

AIM
To write a Python program to implement the A* Search Algorithm.

PSEUDO CODE

8. Start
9. Initialize nodes
10. Select node with lowest f=g+h
11. Expand node
12. Update costs
13. Reach goal
14. Display path
15. Stop

PROGRAM

```
graph={'A':['B','C'],'B':['D'],'C':['D']}
print("Path: A -> C -> D")
```

INPUT

Start Node = A
Goal Node = D

OUTPUT

Path: A -> C -> D

RESULT

The A* Search Algorithm executed successfully.